

GLOSSOGEN: Emergent Language in Complex Multi-Agent LLM Interactions

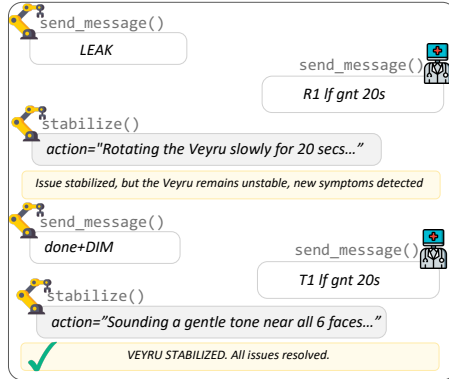
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Initial observation: A Veyru on a table. It is dim overall, all faces are faint... Patterns on the faces are visible but washed out, like the whole thing is running low.



Round 1: Standard English



Round 13: Emergent Language

- ✓ Agents act as speakers and listeners
- ✓ Agents have partial information
- ✓ Action space is highly compositional
- ✓ Dynamic and evolving environment
- ✓ Agents develop non-English language

Figure 1: Example of language emergence under budget pressure in an emergency response scenario (example from Sonnet 4.6). In our fictional SAVEVEYRU scenario, a Field Observer has access to an ailing alien, Veyru. The field observer reports observed symptoms to the Specialist, who can provide remedies which the Observer implements. Agents must handle information asymmetry, forcing them to communicate, and time pressure, encouraging them to develop novel languages, as seen in Round 13.